

Applications: Growth and Decay



Growth models

- 1 A city's population is modeled by $P(t) = 12000(1.035)^t$, where t is in years.
 - a State the initial population and the annual growth rate.
 - b Find the population after 10 years, to the nearest person.
 - 2 An account earns 4% annual interest compounded continuously. How long does it take an investment to double? Give an exact expression and a decimal to two places.
-

Decay models

- 3 Carbon-14 has a half-life of 5730 years: $A(t) = A_0\left(\frac{1}{2}\right)^{t/5730}$.
 - a What fraction of the original carbon-14 remains after 2000 years? Answer to three decimal places.
 - b A sample retains 30% of its carbon-14. Estimate its age to the nearest hundred years.
 - 4 A 200 g sample of a radioactive substance decays to 50 g in 12 hours, following $A(t) = 200e^{kt}$.
 - a Find the decay constant k (4 d.p.).
 - b Find the half-life of the substance.
 - 5 Newton's law of cooling: coffee at 90°C cools in a 20°C room following $T(t) = 20 + 70e^{-0.1t}$ (t in minutes). Find the temperature after 10 minutes, to one decimal place.
-

Logistic growth (an introduction)

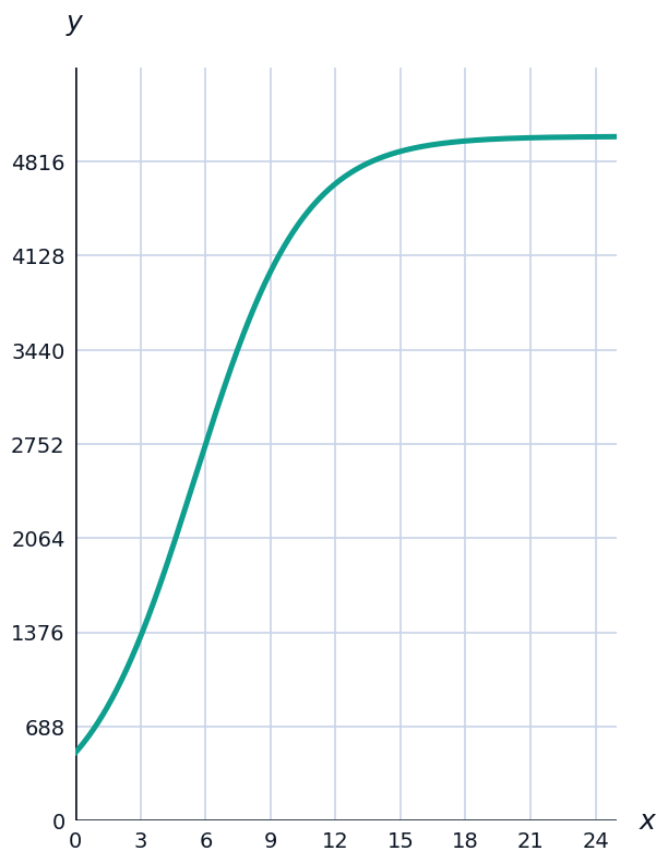
- 6 A lake is stocked with fish, and the population is modeled by $P(t) = \frac{5000}{1 + 9e^{-0.4t}}$, where t is in years.
 - a Find the initial population $P(0)$.
 - b Find the carrying capacity (the value P approaches as $t \rightarrow \infty$).
 - c Find $P(10)$, to the nearest fish.
-

Comparing models

- 7 Two towns start with 10,000 people. Town A grows at a constant rate of 300 people/year (linear); Town B grows at 3% per year (exponential). Find each town's population after 15 years, to the nearest person, and state which is larger.
- 8 A radioactive isotope decays from 80 g to 10 g over 3 half-lives. Find the half-life in terms of the decay time, and write the exponential decay model $A(t) = 80e^{kt}$, finding k if the total elapsed time is 15 years (4 d.p.).

Visualizing the logistic curve

- 9 The fish population model $P(t) = \frac{5000}{1 + 9e^{-0.4t}}$ (from earlier) is graphed below. Use the graph to estimate where the population is growing fastest, then explain why this happens near the inflection point.



Doubling time and half-life formulas

- 10 Derive a general formula for the doubling time T of a quantity growing as $A(t) = A_0e^{kt}$, in terms of k .
- 11 Using the doubling-time formula derived above, find the doubling time for a population growing at a continuous rate of $k = 0.025$ per year, to one decimal place.

Half-life formula derivation

- 12** Derive a general formula for the half-life $T_{1/2}$ of a quantity decaying as $A(t) = A_0 e^{kt}$ (with $k < 0$), in terms of k .
- 13** A radioactive isotope has decay constant $k = -0.023$ per year. Use the formula above to find its half-life, to one decimal place.

Combining growth models with rates

- 14** A savings account earns continuous interest, growing from 5000 to 5512.50 riyals in 2 years. Find the annual interest rate r using $A = Pe^{rt}$, to two decimal places (as a percentage).